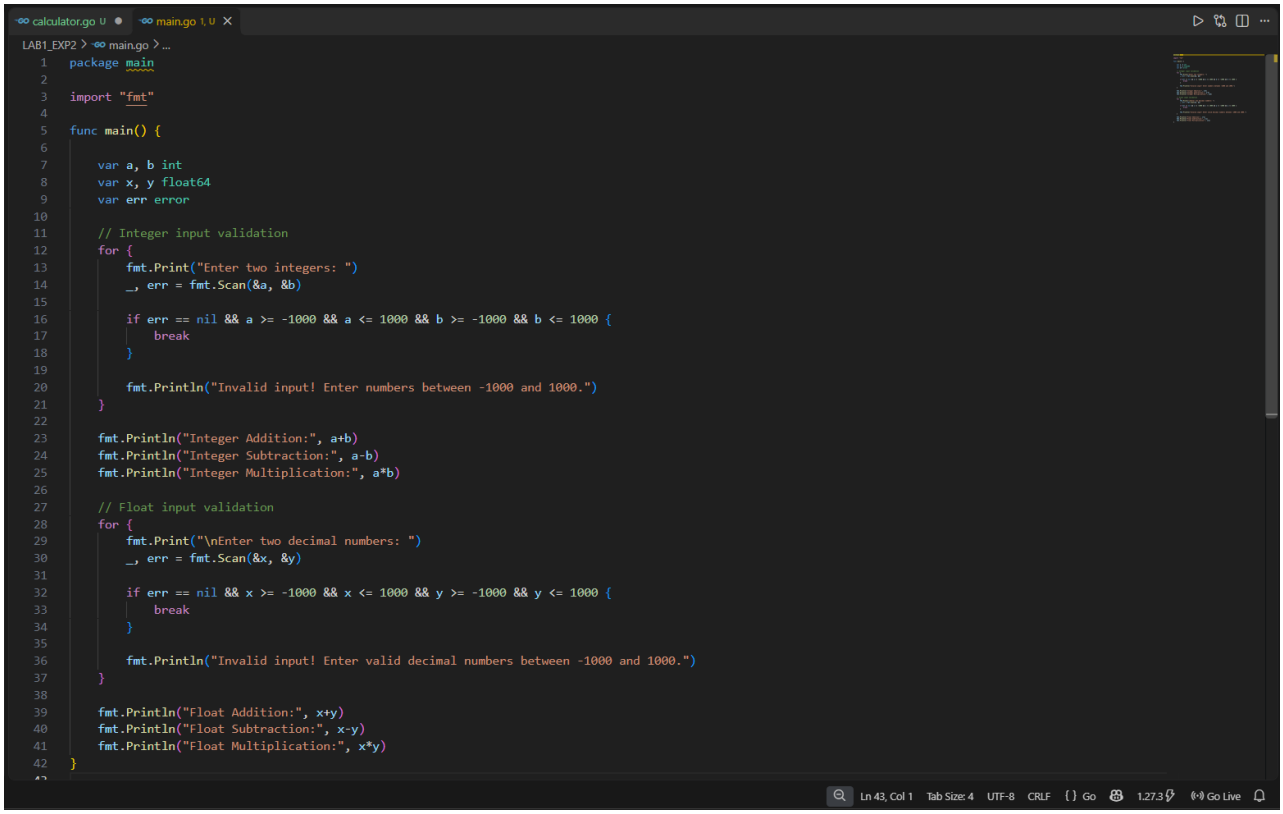
LAB 1

2. Implement a simple calculator program.

    1. Accept numeric input.
    2. Perform addition, subtraction, and multiplication using integer types.
    3. Perform addition, subtraction, and multiplication using floating-point types.
    4. Use if statements and loops to validate the input (reject non-numeric or out-of-range values) and handle different scenarios. (Experiment 1)



```go
package main

import "fmt"

func main() {

    var a, b int
    var x, y float64
    var err error

    // Integer input validation
    for {
        fmt.Print("Enter two integers: ")
        _, err = fmt.Scan(&a, &b)

        if err == nil && a >= -1000 && a <= 1000 && b >= -1000 && b <= 1000 {
            break
        }

        fmt.Println("Invalid input! Enter numbers between -1000 and 1000.")
    }

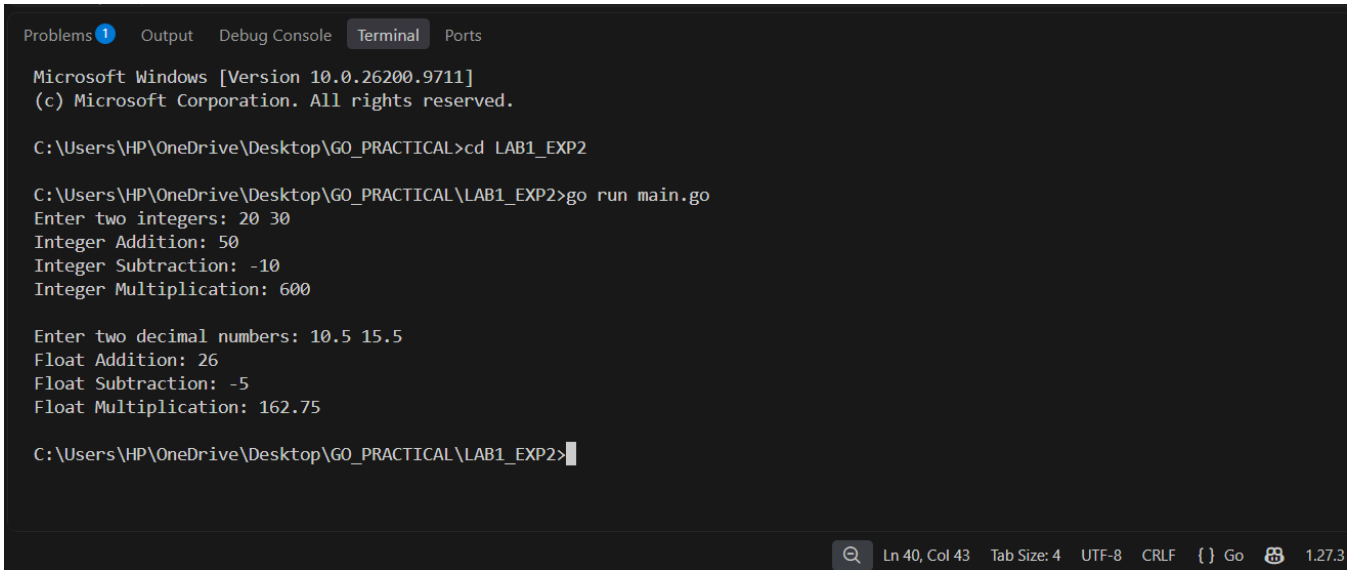
    fmt.Println("Integer Addition:", a+b)
    fmt.Println("Integer Subtraction:", a-b)
    fmt.Println("Integer Multiplication:", a*b)

    // Float input validation
    for {
        fmt.Print("\nEnter two decimal numbers: ")
        _, err = fmt.Scan(&x, &y)

        if err == nil && x >= -1000 && x <= 1000 && y >= -1000 && y <= 1000 {
            break
        }

        fmt.Println("Invalid input! Enter valid decimal numbers between -1000 and 1000.")
    }

    fmt.Println("Float Addition:", x+y)
    fmt.Println("Float Subtraction:", x-y)
    fmt.Println("Float Multiplication:", x*y)
}
```



```
Problems 1  Output  Debug Console  Terminal  Ports

Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL>cd LAB1_EXP2

C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\LAB1_EXP2>go run main.go
Enter two integers: 20 30
Integer Addition: 50
Integer Subtraction: -10
Integer Multiplication: 600

Enter two decimal numbers: 10.5 15.5
Float Addition: 26
Float Subtraction: -5
Float Multiplication: 162.75

C:\Users\HP\OneDrive\Desktop\GO_PRACTICAL\LAB1_EXP2>
```